

inverse matrices, col space, null space

- inverse matrices, col space, rank, null space
- doesn't cover Gaussian Elimination or Row Echelon Form
- Matrices yes describe how space is manipulated $\Rightarrow$ comp graphics
- but also let us solve systems of Equations
- $ax + by + cz = \dots$ is the form
- linear system of equations

$$\begin{array}{ccc} \left[\begin{array}{ccc} a & b & c \\ \vdots & \vdots & \vdots \end{array} \right] & \left[\begin{array}{c} x \\ y \\ z \end{array} \right] & = \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \\ \text{coeff} & \text{var} & \text{constant} \\ A & \vec{x} & = \vec{v} \end{array}$$

$$A \vec{x} = \vec{v}$$

• $A\vec{x}' = \vec{v}'$; A represents some linear transformation.

looking for $\vec{x}'$, which after applying lin transform A represents, lands on $\vec{v}'$.

• does A 's $L()$ squish everything into a lower dimension or not?

• (0 or non 0 determinant)

• $\det(A) \neq 0 \Rightarrow \exists$ 1 vector landing on V

apply transformation in reverse, following where $\vec{v}'$ goes, finds $\vec{x}'$.

• $L_A()$ in reverse is a new $L_A^{-1}()$ called the inverse A^{-1} .

$$\overset{-1}{A} A = \text{no change} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

2nd 1st

←
عربي

Once you have A^{-1} ,

$$\overset{-1}{A} A \vec{x} = \overset{-1}{A} \vec{v}$$

$$\vec{x} = \overset{-1}{A} \vec{v} \quad // \text{ play transformation in reverse \& follow } \vec{v}$$

$\det() \neq 0$, 1 solution bc A^{-1} exists

$\det() = 0$, squish, there is no inverse. you can't unsquish.

A^{-1} doesn't exist.

sol can still exist if $\vec{v}$ lies on the line

• some transformations are squishier than others.

ex: $[3 \times 3] \rightarrow \det = 0 \rightarrow$ squish to plane $\text{rank} = 2$

$[3 \times 3] \rightarrow \det = 0 \rightarrow$ squish to line $\text{rank} = 1$

$\text{rank} = \#$ dimensions
in output of dimension

ex: $[2 \times 2]$ w/ Rank 2 $\Rightarrow$ basis vectors still span all of $\mathbb{R}_2$

• set of all possible outputs of $A\vec{v}$
= col space of matrix

bc col of matrix say were basis vectors
and; $\&$ span of those transformed
basis vectors gives you all possible
outputs.

- $\Rightarrow$ col space = span of cols of matrix
- Rank = dimension in col space.
- Full Rank = no squish in output
- $[0]$ in col space, bc Lin tra must keep origin in place
- For full rank transform, only vector
② origin is the zero vector
- For matrices that aren't full rank,
a bunch of vectors can land on 0.
- Set of vectors that land on origin
 - $\Rightarrow$ null space or kernel of your matrix
 - $\Rightarrow$ space of all vectors that become null

- when A has A^{-1} , you can use it to solve system
- col space: do solutions exist?
- null space: set of all sol could look like?

non square matrices

• how to interpret?

- transformations b/w dimensions exist:

$$\begin{bmatrix} : \\ : \end{bmatrix}_{2D} \rightarrow L() \rightarrow \begin{bmatrix} : \\ : \end{bmatrix}_{3D}$$

- Linear if grid evenly spaced
- & origin still origin

• where $\uparrow$ lands post transform, where $\uparrow$...

$$\begin{matrix} 3 \text{ rows} & \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix} & = 3 \times 2 \\ & \text{2 col} & \end{matrix}$$

span of columns = where all vectors land:

2D plane slicing through origin of 3D space.

bc we only have two equations.

but matrix is still full rank, bc input space & output space have same number of dimensions

3x2 matrix, interpret as mapping 2D to 3D.

2 columns: input space has 2 basis vectors

3 rows: landing spots for each basis vector is described w/ 3 separate coordinates

2x3 matrix, mapping 3D to 2D

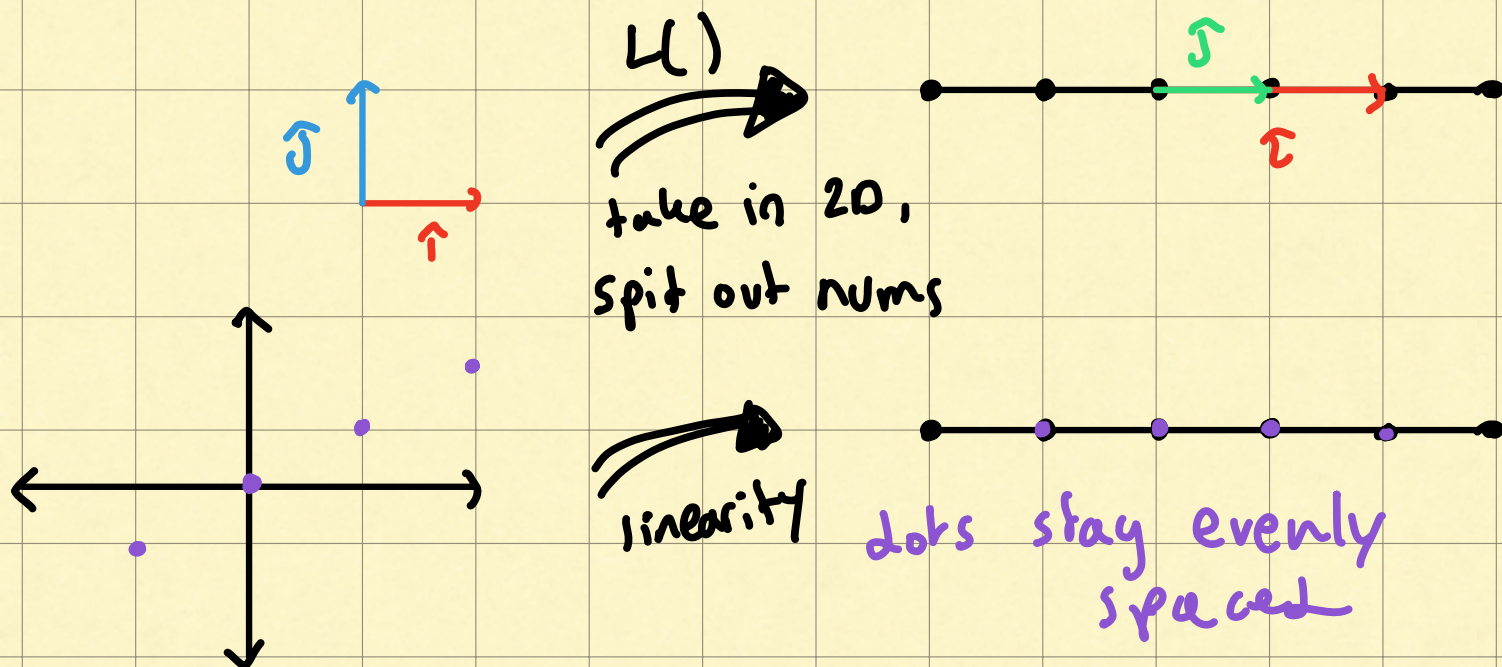
3 col: start w/ 3 basis vectors = 3D

2 row: landing spots of each basis vector described w/ only two coordinates = 2D

from 3D space onto 3D plane

• also 2D to 1D $L()$: $[\cdot] \rightarrow L(\vec{v}) \rightarrow [\cdot]$

• 1D space = number line

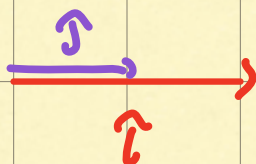


transformation: $[\cdot \cdot]$ 1×2

2 col : input is 2D

1 row: basis vectors land w/ just 1 coord

ex $[2 \ 1]$



dot products

- only w/ linear transformations can we understand the dot product

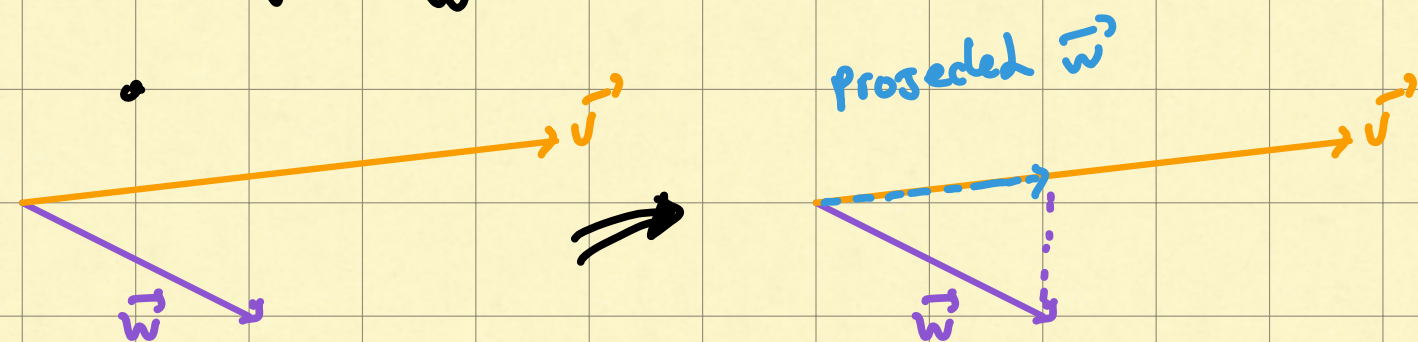
- standard view :

→ given 2 vectors of same dimension

→ dot product = putting up each coordinate, multiply them, adding result

ex: $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} d \\ e \\ f \end{bmatrix} = ad + be + cf$

$\vec{v}$ $\vec{w}$



multiply length projection onto : $\vec{v} \cdot \vec{w} = (|\text{proj}_{\vec{v}} \vec{w}|) \cdot (|\vec{v}|)$

length of $\vec{v}$, = $\vec{v} \cdot \vec{w}$